

Jordan A. Awan
 jaa557@pitt.edu
[Google Scholar](#)

RESEARCH INTERESTS

Data Privacy

Theoretical and applied problems in differential privacy; Statistical inference on privatized data; Theoretical guarantees for synthetic data.

Statistics

Simulation-based inference; Computational statistics, Functional data analysis, Causal inference, U-statistics and kernel tests

Applied Work

Analysis of physiological signals; Acoustic Analyses; Frequency tracking algorithms

EDUCATION

Penn State University, University Park PA

August 2016-May 2020

Doctor of Philosophy, Statistics.

Advised by [Aleksandra Slavković](#) and [Matthew Reimherr](#).

Brandeis University, Waltham MA

Fall 2014-Spring 2016

Master of Arts, Mathematics.

Clarion University of Pennsylvania, Clarion PA

Fall 2011-Spring 2014

Bachelor of Science, Mathematics.

Minors: Computer Science, Honors.

PROFESSIONAL CAREER

University of Pittsburgh, Department of Statistics, Pittsburgh PA

August 2025-present

Assistant Professor.

MITRE

June 2021-September 2025

Differential privacy consultant for MITRE and Census Disclosure Avoidance System team.

Purdue University, Department of Statistics, West Lafayette IN

August 2020-August 2025

Assistant Professor.

Harvard University, Center for Research on Computation and Society (CRCS), Cambridge MA

Summer 2018

Visiting Graduate Student.

Penn State University, Department of Statistics, University Park PA

Summer 2017-Spring 2020

Research Assistant.

Teaching Assistant (Spring 2019). STAT 401.

Lafayette College, Department of Mathematics, Easton PA

Summer 2013

REU participant.

HONORS & AWARDS

Journal of Voice 2025 Best Paper Award Best Paper in the Speech-Language Pathology category	Spring 2026
ASHA Convention Visionary Session, American Speech and Hearing Association Presentation “Estimating Transglottal Airflow Using a Vortex Whistle System and the Phonation Quotient” selected as Visionary	Fall 2024
2024 College of Science Recognition Award, Purdue University Recognized as the recipient of high profile projects	Spring 2024
Faculty/Staff Recognition Award, Purdue Statistics	Spring 2024
2nd place Best Poster Award, Statistics and Optimization in Data Science Workshop, Purdue University	Summer 2023
Outstanding Poster Award, Midwest Machine Learning Symposium	Spring 2023
Regina and Norman Carroll Research Award for 2022, Purdue Statistics Recognized for distinctive contributions to statistical science	Spring 2023
Journal of Voice 2022 Best Paper Award Best Paper in the Speech-Language Pathology category	Spring 2023
The Voice Foundation Best Poster Award	Summer 2021
Penn State 2020 Alumni Dissertation Award	Spring 2020
PSU Statistics 50th Anniversary Best Poster Award	Spring 2018
August and Ruth Homeyer Graduate Fellowship, PSU	Fall 2017-Spring 2018
Best Performance on Applied Qualifying Exam, PSU Statistics	Summer 2017
Stephen B. Brumbach Distinguished Graduate Fellowship, PSU	Fall 2016-Spring 2017
GAANN Fellowship	Fall 2014-Summer 2016
MAA Outstanding Student Poster Award	Winter 2014
Clarion University France-Allison Presentation Award	Fall 2013
MAA Outstanding Student Presentation Award	Summer 2013
Board of Governors Academic Tuition Scholarship	Fall 2011-Spring 2014

SUBMITTED PAPERS & PREPRINTS

“U”:undergraduate student, “G”: graduate student, and “P”: postdoctoral researcher, at the time of the project

1. **Awan, J.**, Barrientos, A. F., Ju, N. “[Statistical Inference for Privatized Data with Unknown Sample Size.](#)” arXiv:2406.06231.
2. **Awan, J.**, Chen, X., Molinari, R. “[Large-Sample Bayesian Approximations for Privatized Data.](#)” arXiv:2604.24817.
3. Chen, Y.-W.^G, Pasupathy, R., **Awan, J.** “[Near-Optimal Private Tests for Simple and MLR Hypotheses.](#)” arXiv:2601.21959.
4. Chen, Y.-W.^G, Sanghi, P.^U, **Awan, J.** “[Particle Filter for Bayesian Inference on Privatized Data.](#)” arXiv:2505.00877.
5. Cho, Y.^G, **Awan, J.** “[Beyond Data Splitting: Full Conformal Prediction by Differential Privacy.](#)” arXiv:2603.07522. **2026 ASA Student Paper Award, Statistical Learning and Data Science Section.**
6. Cho, Y.^G, **Awan, J.** “[Formal Privacy Guarantees with Invariant Statistics.](#)” arXiv:2410.17468.

7. Eng, K.^G, **Awan, J.**, Ju, N., Rao, V., Gong, R. “dapper: Data Augmentation for Private Posterior Estimation in R.” arXiv:2412.14503.
8. Miglioli, C.^P, **Awan, J.** “Incomplete U-Statistics of Equireplicate Designs: Berry–Esseen Bound and Efficient Construction.” arXiv:2510.20755.
9. Moosavi, B.^G, **Awan, J.** “Public Good Provision under Locally Private Signals.” arXiv:2606.24013.
10. Ohnishi, Y.^G, **Awan, J.** “Differentially Private Covariate Balancing Causal Inference.” arXiv:2410.14789.
11. Wang, Z.^G, Chang, A.^G, **Awan, J.** “Optimal Debiased Inference on Privatized Data via Indirect Estimation and Parametric Bootstrap.” arXiv:2507.10746.

REFEREED PUBLICATIONS

1. Awan, S., **Awan, J.**, McKenna, V., Gifford, S., Burns, E., Sun, Z.^G, Chen, J., Eddins, D. (2026+) “Effect of Vowel Type on Aerodynamic Measures of Estimated Subglottal Pressure and Airflow.” *Journal of Speech, Language, and Hearing Research*. Accepted.
2. Awan, S., **Awan, J.** (2025+) “Comparison of Methods of Eliciting Vital Capacity: Forced vs. Slow Vital Capacity.” *Journal of Voice*. Accepted and available online.
3. Awan, S., **Awan, J.**, Bonilha, H., Gillespie, A., McKenna, V., Chen, J., Eddins, D. (2025+) “Using a Vortex Whistle System to Estimate Phonatory Airflow via the Phonation Quotient.” *Journal of Voice*. Accepted and available online.
4. **Awan, J.**, Cai, Z. (2025) “One Step to Efficient Synthetic Data.” *Statistica Sinica*. Volume 35, Pages 531-569.
5. **Awan, J.**, Edwards, A., Bartholomew, P., Sillers, A. (2025) “Best Linear Unbiased Estimate from Privatized Histograms.” *Journal of Machine Learning Research*. Volume 26, Issue 174, Pages 1-41.
6. **Awan, J.**, Wang, Y.^U (2025) “Differentially Private Kolmogorov-Smirnov-Type Tests.” *Electronic Journal of Statistics*. Volume 19, No. 1, Pages 718-744.
7. **Awan, J.**, Wang, Z.^G (2025) “Simulation-Based Finite-Sample Inference for Privatized Data.” *Journal of the American Statistical Association*. Pages 1-14.
8. Awan, S., Shaikh, M., **Awan, J.**, Abdalla, I., Lim, K., Misono, S., (2025) “Smartphone Recordings are Comparable to ‘Gold Standard’ Recordings for Acoustic Measurements of Voice.” *Journal of Voice*. Volume 39, Issue 4, Pages 1019-1032. **Best Paper Award**.
9. Chen, Y.-W.^G, Pasupathy, R., **Awan, J.** (2025) “Optimal Survey Design for Private Mean Estimation.” *Proceedings of the 42nd International Conference on Machine Learning*. Pages 8856–8870.
10. Feinstein, H.^G, Daşdöğen, Ü.^G, **Awan, J.**, Awan, S., Verdolini Abbott, K. (2025) “Comparative Analysis of Two Methods of Perceptual Voice Assessment.” *Journal of Voice*. Volume 39, Issue 4, Pages 1065-1075.
11. Ohnishi, Y.^G, **Awan, J.** (2025) “Locally Private Causal Inference for Randomized Experiments.” *Journal of Machine Learning Research*. Volume 26, No. 14, Pages 1-40.
12. Wang, Z.^G, Cheng, G., **Awan, J.** (2025) “Differentially Private Bootstrap: New Privacy Analysis and Inference Strategies.” *Journal of Machine Learning Research*. Volume 26, Issue 257, Pages 1-57.
13. **Awan, J.**, Bernardi, O. (2024) “Tutte Polynomials for Regular Oriented Matroids.” *Discrete Mathematics*. Volume 347, Number 1.
14. **Awan, J.**, Ramasethu, A.^G (2024) “Optimizing Noise for f -Differential Privacy via Anti-Concentration and Stochastic Dominance.” *Journal of Machine Learning Research*. Volume 25, Number 351, Pages 1-32.
15. Kang, T.^G, Kim, S.^G, Sohn, J.^G, **Awan, J.** (2024) “Differentially Private Topological Data Analysis.” *Journal of Machine Learning Research*. Volume 25, No. 189, Pages 1-42.
16. **Awan, J.**, Rao, V. (2023) “Privacy-Aware Rejection Sampling.” *Journal of Machine Learning Research*. Volume 24, No. 74, Pages 1-32.
17. **Awan, J.**, Vadhan, S. (2023) “Canonical Noise Distributions and Private Hypothesis Tests.” *Annals of Statistics*. Volume 51, Number 2, Pages 547-572.
18. **Awan, J.**, Dong, J. (2022) “Log-Concave and Multivariate Canonical Noise Distributions for Differential Privacy.” *Advances in Neural Information Processing Systems* 36, 34229-34240.

19. **Awan, J.**, Frechette, C., Li, Y., McMahon, E. (2022) “[Demicaps in \$AG\(4, 3\)\$ and their Relation to Maximal Cap Partitions.](#)” *Graphs and Combinatorics*. Volume 83, No. 193.
20. Awan, S., **Awan, J.** (2022) “[Use of a Vortex Whistle for Measures of Respiratory Capacity.](#)” *Journal of Voice*. Volume 36, Issue 5, Pages 630-636. **Best Paper Award.**
21. Ju, N., **Awan, J.**, Gong, R., Rao, V. (2022) “[Data Augmentation MCMC for Bayesian Inference from Privatized Data.](#)” *Advances in Neural Information Processing Systems 36*, 12732-12743.
22. Li, A.^G, **Awan, J.**, Chen, J., Eddins, D., Awan, S. (2022) “[Enhancing the Vortex Whistle for Measures of Respiratory Capacity via CFD and CAA.](#)” *Journal of Biomechanical Engineering*. Volume 144, Issue 11.
23. Li, A.^G, Chen, J., **Awan, J.**, Eddins, D., Awan, S. (2022) “[Performance Analysis and Parametric Study of Vortex Whistle.](#)” *Proceedings of the ASME 2022 Fluids Engineering Division Summer Meeting. Volume 1: Fluid Applications and Systems (FASTC); Fluid Measurement and Instrumentation (FMITC); Fluid Mechanics (FMTTC)*. Toronto, Ontario, Canada. August 3–5, 2022. V001T01A018. ASME.
24. **Awan, J.**, Slavković, A. (2021) “[Structure and Sensitivity in Differential Privacy: Comparing \$K\$ -Norm Mechanisms.](#)” *Journal of the American Statistical Association*. Volume 116, Number 534, 935-954.
25. **Awan, J.**, Bernardi, O. (2020) “[Tutte Polynomials for Directed Graphs.](#)” *Journal of Combinatorial Theory, Series B*. Volume 140, 192-247.
26. **Awan, J.**, Slavković, A. (2020) “[Differentially Private Inference for Binomial Data.](#)” *Journal of Privacy and Confidentiality*. Volume 10, No. 1.
27. Awan, S., **Awan, J.** (2020) “[A Two-Stage Cepstral Analysis Procedure for the Classification of Rough Voices.](#)” *Journal of Voice*. Volume 34, Issue 1, 9-19.
28. **Awan, J.**, Kenney, A., Reimherr, M., Slavković A. (2019) “[Benefits and Pitfalls of the Exponential Mechanism with Applications to Hilbert Spaces and Functional PCA.](#)” *Proceedings of the 36th International Conference on Machine Learning*, 97:374-384.
29. **Awan, J.**, Slavković, A. (2018) “[Differentially Private Uniformly Most Powerful Tests for Binomial Data.](#)” *Advances in Neural Information Processing Systems 32*, 4208-4218.
30. Reimherr, M., **Awan, J.** (2019) “[KNG: The \$K\$ -Norm Gradient Mechanism.](#)” *Advances in Neural Information Processing Systems 33*. 10208-10219.
31. Reimherr, M., **Awan, J.** (2019) “[Elliptical Perturbations for Differential Privacy.](#)” *Advances in Neural Information Processing Systems 33*. 10185-10196.
32. Gaskill, C., **Awan, J.**, Watts, C., Awan, S. (2017) “[Acoustic and Perceptual Classification of Within-sample Normal, Intermittently Dysphonic, and Consistently Dysphonic Voice Types.](#)” *Journal of Voice*, Volume 31, Issue 2, Pages 218-228.
33. Awan, S., **Awan, J.** (2013) “[The Effect of Gender on Measures of Electrolaryngographic Contact Quotient.](#)” *Journal of Voice*, Volume 27, Issue 4, 433-440.

BOOK CHAPTERS

1. **Awan, J.**, Gong, R. (2024). “[Statistical Inference and Differential Privacy.](#)” In Drechsler, J., Kifer, D., Reiter, J., & Slavković, A. (Eds.), *Handbook of Sharing Confidential Data: Differential Privacy, Secure Multiparty Computation, and Synthetic Data*. Chapman and Hall/CRC.

OTHER PUBLICATIONS

1. **Awan, J.** (2024). “[Here’s How Machine Learning can Violate your Privacy.](#)” *The Conversation*. May 23, 2024.
2. Habib, S., Pires, B., Benedetto, G., Rodriguez, R., **Awan, J.**, Stanley, J., Totty, E., Germinario, G., & Stevenson, R. (2023). “[Automated Synthetic Data Validation: Applying Noise Injection for Disclosure Avoidance.](#)” *Joint Statistical Meetings (JSM)*, Toronto, Canada.

3. **Awan, J.**, Reimherr, M., Slavković, A. (2020). “Formal Privacy for Modern Nonparametric Statistics.” *CHANCE* 33, No. 4. 43-49.
4. Awan, S., **Awan, J.**, Watts, C., S. Gaskill, C. (2018). “Response to Aichinger and Kubin Re: Letter to the Editor “Acoustic and Perceptual Classification of Within-Sample Normal, Intermittently Dysphonic, and Consistently Dysphonic Voice Types”.” *Journal of Voice*. Issue 32, No. 3, 383-384.

GRANTS

NIH R01: Deconstructing Voice Therapy: Towards Enhanced Communication Outcomes, Co-I **2025-2030**

Award number: 1R01DC022026-01A1

Co-investigator.

PI: Dr. Amanda Gillespie.

Co-Is: Drs. Kris Tjaden, Eva van Leer, Anthony Law, Adam Klein

\$2,821,462 for 5 years.

Directly responsible for $\approx$ \$300,000.

NIH R01: Vital Capacity & Airflow Measurement for Voice Evaluation: A Vortex Whistle System, MPI **2023-2028**

Award number: R01 DC020799-01A1

One of 4 MPIs, along with Drs. Shaheen Awan, Jun Chen, and Amanda Gillespie.

Co-Is: Drs. David Eddins and Victoria McKenna

\$3,129,418 for 5 years.

Directly responsible for $\approx$ \$450,000.

NSF SES: Simulation-Based Inference for Differential Privacy, PI **2022-2027**

Current award number: SES 2610910

Prior award number to Purdue: SES 2150615

Principal investigator

Co-PI: Dr. Roberto Molinari

\$450,000.

PENDING PATENTS

261481 — 2085.227.PR3 - Multi-Channel High-Flow Vortex Whistle Spirometer **2026**

Patent application filed

With Shaheen Awan, Jun Chen, David Eddins, and Zheming Sun

256588 — 2085.227.PR - Low Flow Vortex Whistle Airflow Measurement System **2025**

Provisional patent is pending

With Shaheen Awan, Jun Chen, David Eddins, and Zheming Sun

INVITED CONFERENCE PRESENTATIONS

ENAR 2026, Indianapolis, IN **March 2026**

Simulation-Based, Finite-Sample Inference for Privatized Data

Joint Statistical Meetings, Nashville, TN **August 2025**

Best Linear Unbiased Estimate from Privatized Histograms

Joint Statistical Meetings, Portland, OR **August 2024**

Panelist: Evaluating Statistical Disclosure Control Techniques based on the Risk and Utility of Privacy-Protected Data

25th Annual CERIAS Security Symposium, Purdue University, West Lafayette, IN	April 2024
Valid Statistical Inference on Privatized Data	
Statistical Learning and Differential Privacy, Bath U.K. (online)	September 2022
Data Augmentation MCMC for Bayesian Inference from Privatized Data	
Workshop on the Analysis of Census Noisy Measurement Files and Differential Privacy, Rutgers University	April 2022
Posterior Inference on Privatized Data via Data Augmentation MCMC	
Computational & Methodological Statistics Meeting, Online	December 2021
Canonical noise distributions and private hypothesis tests	
Virtual Symposium on Data Privacy, ASA Nevada Chapter	September 2021
Canonical noise distributions and private hypothesis tests	
Computational & Methodological Statistics Meeting in Pisa, Italy	December 2018
Differentially private UMP hypothesis tests for Bernoulli data	
Statistical Society of Canada Annual Meeting, McGill University, Montreal Canada	June 2018
Optimizing finite sample performance under differential privacy	

INVITED DEPARTMENTAL and INDUSTRY SEMINARS

Virginia Tech, Department of Statistics, Blacksburg, VA	March 2026
Beyond Data Splitting: Full Conformal Prediction by Differential Privacy	
University of Pittsburgh, Machine Learning Seminar, Department of Mathematics, Pittsburgh, PA	February 2026
Differentially Private Bootstrap: New Privacy Analysis and Inference Strategies	
University of Pittsburgh, Department of Biostatistics, Pittsburgh, PA	January 2026
Simulation-Based, Finite-Sample Inference for Privatized Data	
Capital One, McLean VA (virtual)	January 2026
Best Linear Unbiased Estimate from Privatized Contingency Tables	
National Institute for Research in Digital Science and Technology (Inria), Montpellier, France (virtual)	November 2025
Differentially Private Bootstrap: New Privacy Analysis and Inference Strategies	
University of Pittsburgh, Department of Statistics, Pittsburgh, PA	December 2024
Simulation-Based, Finite-Sample Inference for Privatized Data	
Auburn University, Department of Mathematics and Statistics, Auburn, AL	April 2024
Simulation-Based, Finite-Sample Inference for Privatized Data	
Air Force Institute of Technology, Department of Mathematics and Statistics, Wright-Patterson Air Force Base, OH	January 2023
Bayesian Inference on Privatized Data	
Auburn University, Statistics and Data Science Seminar, Online	September 2022
Bayesian Inference from Privatized Data	
Michigan State University, Department of Statistics, Online	November 2021
Canonical noise and private hypothesis tests	
University of Wisconsin-Madison, Department of Statistics, Madison WI	February 2020

Differentially private inference for binomial data	
Lafayette College, Department of Mathematics, Easton PA	February 2020
Differentially private inference for binomial data	
George Mason University, Department of Statistics, Fairfax VA	February 2020
Differentially private inference for binomial data	
Bucknell University, Department of Mathematics, Lewisburg PA	January 2020
Differentially private inference for binomial data	
Purdue University, Department of Statistics, West Lafayette IN	January 2020
Differentially private inference for binomial data	
Stochastic Modeling and Computational Statistics Seminar at Penn State, University Park PA	February 2018
Structure and sensitivity in DP: comparing K -korm mechanisms	
MIT Combinatorics Seminar, Cambridge MA	April 2016
Tutte polynomials for directed graphs and oriented matroids	
Brandeis Combinatorics Seminar, Waltham MA	January 2016
Tutte polynomials for directed graphs and oriented matroids	
Clarion University Honors Presentations, Clarion PA	Fall 2013
Results on demicaps in $AG(4, 3)$	

CONTRIBUTED CONFERENCE PRESENTATIONS

Joint Statistical Meetings, Toronto Canada	August 2023
Simulation-Based Inference for Privatized Data	
Joint Statistical Meetings, Washington D.C.	August 2022
Posterior Inference on Privatized Data via Data Augmentation MCMC	
Privacy in Machine Learning, Virtual NeurIPS Workshop	December 2021
Canonical noise distributions and private hypothesis tests	
Privacy in Machine Learning, Virtual NeurIPS Workshop	December 2021
Privacy-aware rejection sampling	
Privacy Preserving Machine Learning, Virtual ACM CCS Workshop	November 2021
Canonical noise and private hypothesis tests	
2021 Joint Statistical Meetings, Online	August 2021
Approximate co-sufficient sampling with applications to goodness of fit tests and synthetic data	
2020 Joint Statistical Meetings, Online	August 2020
KNG: The K-norm gradient mechanism	
2019 Joint Statistical Meetings, Denver CO	July 2019
Benefits and pitfalls of the exponential mechanism	
36th International Conference Machine Learning, Long Beach CA	June 2019
Benefits and pitfalls of the exponential mechanism	
Simons Institute for the Theory of Computing, Berkeley CA	April 2019
Differentially private UMP hypothesis tests for Bernoulli data	
2018 Joint Statistical Meetings, Vancouver Canada	July 2018

Optimizing finite sample performance under differential privacy

Mathematical Foundations of Data Privacy, Banff International Research Station (BIRS), Banff Canada	May 2018
Structure and sensitivity in DP: comparing K -norm mechanisms	
Pi Mu Epsilon Conference, Youngstown OH	Spring 2013
Maximal caps and substructures in $AG(3, 3)$	
Joint Math Meetings, Baltimore MD	Winter 2013
Maximal caps and substructures in $AG(3, 3)$	
Mathfest Conference, Hartford CT	Summer 2013
Maximal caps and substructures in $AG(4, 3)$	

POSTERS

Thirty-Sixth Conference on Neural Information Processing Systems, New Orleans, LA (online)	November 2022
Log-Concave and Multivariate Canonical Noise Distributions for Differential Privacy	
Thirty-Sixth Conference on Neural Information Processing Systems, New Orleans, LA (online)	November 2022
Data Augmentation MCMC for Bayesian Inference from Privatized Data	
Privacy in Machine Learning, Virtual NeurIPS Workshop	December 2021
Canonical noise distributions and private hypothesis tests	
Privacy in Machine Learning, Virtual NeurIPS Workshop	December 2021
Privacy-aware rejection sampling	
Privacy Preserving Machine Learning, Virtual ACM CCS Workshop	November 2021
Canonical noise and private hypothesis tests	
Privacy Preserving Machine Learning, Virtual ACM CCS Workshop	November 2021
Privacy-aware rejection sampling	
Thirty-Third Conference on Neural Information Processing Systems, Vancouver Canada	December 2019
Elliptical perturbations for differential privacy	
Thirty-Third Conference on Neural Information Processing Systems, Vancouver Canada	December 2019
K -Norm gradient mechanism for private empirical risk minimization	
36th International Conference on Machine Learning, Long Beach CA	Summer 2019
Benefits and pitfalls of the exponential mechanism	
Thirty-second Conference on Neural Information Processing Systems, Montreal Canada	December 2018
Differentially private uniformly most powerful tests for binomial data	
Theory and Practice of Differential Privacy in 25th ACM Conference on Computer and Communications Security, Toronto Canada	October 2018
Differentially private uniformly most powerful tests for binomial data	
50th Anniversary Conference at Penn State Department of Statistics, University Park PA	May 2018

Optimizing finite sample performance under differential privacy

Rao Prize Conference at Penn State, University Park PA

May 2017

Maximum likelihood estimation with differential privacy

Joint Math Meetings, Baltimore MD

Winter 2014

REU results on maximal caps and substructures in $AG(4, 3)$

OTHER PRESENTATIONS

Lilly Purdue Statistics Seminar, Eli Lilly and Company, Indianapolis IN

Spring 2024

Statistical Inference with Differential Privacy

Open DP Community Workshop

Summer 2020

Lightning talk on Binomial inference under differential privacy

Penn State Statistics Graduate Student Association Workshop

Fall 2018

Introduction to differential privacy

Center for Research on Computation and Society, Harvard University

Summer 2018

Introduction to differential privacy

Penn State Statistics Graduate Student Association Workshop

Fall 2017

Introduction to differential privacy

Penn State DS 300: Privacy and Security for Data Sciences

Fall 2017

Introduction to differential privacy

Brandeis Mathematics Graduate Student Seminar

Fall 2015

A proof of the 5 color theorem

Brandeis Combinatorics Seminar

Spring 2015

Topics in matroid representability

Brandeis Mathematics Graduate Student Seminar

Spring 2015

Topics regarding the Tutte polynomial

Pi Mu Epsilon Conference, Youngstown OH

Spring 2013

A solution for the 2013 COMAP MCM problem A

Clarion University High School Mathematics Competition

Fall 2012

Mental math algorithms with proofs and examples

Cumberland Valley Math Modeling Challenge at Shippensburg University

Fall 2011

A model to predict the economic impacts of different voting systems

TEACHING EXPERIENCE

University of Pittsburgh, Department of Statistics, Instructor

Fall 2025-present

STAT 3692: Differential Privacy, Fall 2025

STAT 1261/2260: Introduction to Data Science, Spring 2026

Purdue University, Department of Statistics, Instructor

Fall 2020-Spring 2025

CS/STAT 242: Introduction to Data Science, Spring 2024

STAT 598: Differential Privacy, Fall 2022, Spring 2025

MA/STAT 519: Probability Theory, Fall 2021, Spring 2023, Spring 2025

STAT 692: Research Seminar, Fall 2021, Spring 2022

STAT 417: Statistical Theory, Fall 2020 (2 sections online), Fall 2022

Pennsylvania State University, Department of Statistics, Instructor	Spring 2019
Introduction to Probability and Statistics with R for Engineers	
Brandeis University, Department of Mathematics, Instructor	Fall 2015, Spring 2016
Integral Calculus	
Brandeis University, Department of Mathematics, Grader	Fall 2014, Spring 2015
Multivariate Calculus, Linear Algebra	
Brandeis University, Department of Mathematics, Tutor	Fall 2014, Spring 2015
Pre-Calculus, Calculus I & II	
Clarion University, Department of Academic Enrichment, Tutor	Fall 2011-Spring 2014
Finite Mathematics, Pre-Calculus, Calculus I & II, Linear Algebra	

SOFTWARE DEVELOPMENT

Canonical Noise Mechanism in OpenDP	2022-2024
Implementation of the canonical noise distributions in “Canonical Noise Distributions and Private Hypothesis Tests” within the OpenDP framework. In collaboration with Aishwarya Ramasethu, Yu-Ju Ku, and Michael Shoemate.	
SimBaRepro: Simulation-Based Finite Sample Inference via Repro Samples	Summer 2025
R package implementation of the method in “Simulation-Based Finite Sample Inference for Privatized Data.” In collaboration with Xinlong Du and Zhanyu Wang. Available on CRAN.	
dapper: Data Augmentation for Private Posterior Estimation in R	Summer 2024
R package implementation of the method in “Data Augmentation MCMC for Bayesian Inference from Privatized Data.” In collaboration with Kevin Eng and Drs. Ruobin Gong, Nianqiao Ju, and Vinayak Rao. Available on CRAN.	
binomialDP: Differentially Private Inference for Binomial Data	Summer 2020
R package implementation of UMP tests and UMA confidence intervals for Binomial test statistics under differential privacy. In collaboration with Tran Tran and Dr. Aleksandra Slavković.	

SERVICE

Journal Referee	
Annals of Statistics, Journal of the Royal Statistical Society Series B, Journal of the American Statistical Association, Neural Information Processing Systems, International Conference on Machine Learning, Journal of Privacy and Confidentiality, Journal of Computational and Graphical Statistics, Statistica Sinica, among others	
Session Organizer, CFE-CMStatistics 2026	Spring 2026- present
Organized a session on statistical inference on privatized data	
Poster Judge, Pittsburgh Chapter of American Statistical Association	Spring 2026
Rated posters by graduate students	
Co-Guest-Editor, Statistical Theory and Related Methods	Fall 2025-present
Special issue on differential privacy	
Colloquium Chair, Pitt Statistics	Fall 2025-Spring 2026
Organized the University of Pittsburgh Statistics seminar, invited speakers	
Program Committee Member, Theory and Practice of Differential Privacy	2021, 2024, 2026
Reviewed submissions and helped organize the workshop	
Pre-Examiner for PhD Thesis, University of Helsinki	Summer 2025

Reviewed the PhD thesis of Ossi Räisä	
Session Organizer, Bayesian, Fiducial, & Frequentist Conference (BFF9)	Spring 2025
Organized a session on statistical inference on privatized data	
PhD Qualifying Exam Committee, Purdue Statistics	Spring 2025
Wrote exam questions and helped to administer the exam	
Faculty Mentor, Science Scholars Program, Purdue	Fall 2024-Spring 2025
Mentored an undergraduate student from an under-represented minority background	
Program Committee, AAAI	Fall 2024
Reviewed articles for the conference	
Program Committee, Fairness, Accountability, and Transparency (FAccT)	Spring 2023
Reviewed submissions for the workshop	
Pre-Examiner for PhD Thesis, Aalto University	Spring 2023
Reviewed PhD thesis of Joonas Jälkö	
Colloquium Chair, Purdue Statistics	Fall 2021-Spring 2022
Organized the Purdue Department of Statistics seminar, invited speakers	
Diversity and Inclusion Committee, Purdue Statistics	2021-2025
Program Committee, NeurIPS Workshop: Privacy and Machine Learning	Fall 2021
Reviewed submissions and helped organize the workshop	
Program Committee, CCS Workshop: Privacy Preserving Machine Learning	Fall 2021
Reviewed submissions and helped organize the workshop	
Graduate Student Admissions, Purdue Statistics	2021-2025
Distinguished Theme Seminar Series, Purdue Statistics	Fall 2020-Fall 2023
Member of the organizing committee (Spring 2021-Fall 2021)	
Seminar Moderator (Fall 2020, Fall 2021)	
Hiring Committee, Purdue Statistics	Fall 2020 - present
Assistant Professor Search; Escort for interviewees (Fall 2020-Spring 2021)	
Assistant and Associate Professor Search (Fall 2021-Spring 2022)	
Dream Hire Search (Fall 2023)	
Assistant Professor Search (Fall 2024-Spring 2025)	

POSTDOCTORAL ADVISOR

Cesare Miglioli, Postdoctoral Researcher	August 2024-August 2026
Next job: Postdoctoral Researcher in Statistics for Psychology, Section of Psychology, Faculty of Psychology and Educational Sciences (FPSE), University of Geneva	

THESIS ADVISOR

Behrooz Moosavi, PhD Student in Economics, Pitt	Summer 2026-present
Co-advised by Arie Beresteanu	
Yu Wei Chen, PhD Student in Statistics, Purdue	Spring 2024-present
Co-advised by Raghu Pasupathy	
Young Hyun Cho, PhD Student in Statistics, Purdue	Spring 2023-Spring 2026

Co-advised by Will Wei Sun

Note that due to an administrative technicality, I am not listed as co-chair on the thesis

First and current job: Postdoctoral Researcher, Department of Statistics, Harvard University

Yuki Ohnishi, PhD Student in Statistics, Purdue

Summer 2022-Spring 2023

Co-advised by Arman Sabbaghi

First job: Postdoctoral Associate, Department of Biostatistics, Yale School of Public Health

Current job: Assistant Professor, Department of Statistics, University of Florida

Zhanyu Wang, PhD Student in Statistics, Purdue

Fall 2021-Fall 2023

Co-advised by Guang Cheng

First and current job: Research Scientist, Meta

OTHER SUPERVISED STUDENTS

Malhar Dave, M.S. Student, Pitt

Summer 2026-present

Private inference via monotone functions

Feier Chang, PhD Student in Biostatistics, Pitt

Spring 2026-present

Differentially private and federated learning for generalized linear models

Haotian Chen, PhD Student, Purdue

Fall 2025-present

Functional data analysis and Gaussian processes for differential privacy

Arin Chang, PhD Student, Purdue

Fall 2024-present

Indirect inference and parametric bootstrap for privatized data

Hamiltonian MCMC for Bayesian inference on privatized data

Pranav Sanghi, Undergraduate Student, Purdue

Spring 2024-present

Simulations for Bayesian inference on privatized data

Leo Navarro, Undergraduate Student, Purdue

Summer 2024-Spring 2025

Optimize Bayesian inference on privatized data

Kefan Gu, Undergraduate Student, Purdue

Summer 2024-Fall 2024

Optimize Bayesian inference on privatized data

Xinlong Du, M.S. Student, Purdue

Spring 2024-Summer 2025

R Package development for simulation-based inference

Samuel Forfang, Undergraduate Student, Purdue

Spring 2024

R Package development for simulation-based inference

Aidan Davis, Undergraduate Student, Purdue

Spring 2024

R Package development for simulation-based inference

Andrew Liu, M.S. Student, Purdue

Fall 2023-Spring 2025

Optimize the subsample and aggregate method for confidence intervals

Aishwarya Ramasethu, M.S. Student, Purdue

Fall 2022-Spring 2023

Research discrete canonical noise distributions and implement binomialDP in OpenDP

Yu-Ju Ku, M.S. Student, Purdue

Summer 2022-Spring 2023

Implement binomialDP in OpenDP

Burla Ondes, PhD Student in I.E., Purdue

Summer 2022

Investigated the EM algorithm to analyze privatized data	
Taegyu Kang, PhD Student, Purdue	Spring 2022-Summer 2024
Differentially private topological data analysis (group project)	
Sehwan Kim, PhD Student, Purdue	Spring 2022-Summer 2024
Differentially private topological data analysis (group project)	
Formalizing semi-privacy (group project)	
Jinwon Sohn, PhD Student, Purdue	Spring 2022-Summer 2024
Differentially private topological data analysis (group project)	
Yue Wang, Undergraduate Student, Purdue	Fall 2021-Summer 2022
Simulation study to compare differentially private hypothesis tests	
Vishnu Suresh (co-chair), M.S. Student, Purdue	Spring 2021-Summer 2021
Exploring research topics in differential privacy	
Jacob Moore, Undergraduate Student, Purdue	Spring 2021-Summer 2021
Developing an R package for approximate conditional sampling	

PHD COMMITTEE

Arin Chang, PhD Student (Statistics, Purdue)	Fall 2025-present
Advised by Vinayak Rao	
Haotian Chen, PhD Student (Statistics, Purdue)	Fall 2025-present
Advised by Qifan Song	
Shubha Sankar Banerjee, PhD Student (Statistics, Pitt)	Summer 2025-present
Advised by Zhao Ren	
Keon Wong Hur, PhD Student (Mechanical Engineering, Purdue)	Spring 2025-present
Advised by Jun Chen	
Hyunwoo Chung, PhD Student (Statistics, Purdue)	Spring 2024-present
Advised by Fei Xue	
Qian Zhang, PhD Student (Statistics, Purdue)	Fall 2022-present
Advised by Faming Liang	
Rajdeep Haldar, PhD Student (Statistics, Purdue)	Spring 2022-Spring 2026
Advised by Qifan Song	
Yi Chu, PhD Student (Statistics, Purdue)	Summer 2022-Summer 2025
Advised by Raghu Pasupathy	
Jiajun Liang, PhD Student (Statistics, Purdue)	Spring 2022-Fall 2023
Advised by Qifan Song	
Xinyi Pei, PhD Student (Statistics, Purdue)	Spring 2021-Spring 2025
Advised by Vinayak Rao	

MS COMMITTEE

Malhar Dave, M.S., Pitt (chair)	Spring 2026-present
These project on differential privacy	
Zheming Sun, M.S. (Mechanical Engineering), Purdue	Fall 2024-present
Andrew Liu, M.S., Purdue (chair)	Fall 2023-Spring 2025

Chair of the MS advisory committee. Reading course in differential privacy

Ian Hunter, M.S., Purdue **Spring 2023-Spring 2024**

Nicholas Rosenorn, M.S. (CS & Statistics), Purdue (chair) **Spring 2023-Fall 2024**

Madison Dunn, M.S., Purdue (chair) **Fall 2022-Spring 2024**

Aishwarya Ramasethu, M.S., Purdue **Fall 2022-Spring 2024**

Chair of the MS advisory committee. Reading course in differential privacy

Yu-Ju Ku, M.S. (CS & Statistics), Purdue (chair) **Summer 2022-Spring 2023**

Quisi Zhang, M.S., Purdue **Spring 2022-Spring 2023**

Qi Zhong, M.S., Purdue **Spring 2022-Fall 2022**

Pratiksha Agrawal, M.S., Purdue **Spring 2022-Spring 2023**

Yi-Min Yang, M.S., Purdue **Fall 2021-Spring 2023**

Vidhi Jain, M.S., Purdue **Fall 2021-Fall 2022**

Yu-Wen Wang, M.S., Purdue **Fall 2021-Spring 2023**

Yi-Ting Hung, M.S., Purdue (co-chair) **Fall 2021-Spring 2023**

John Lambrecht, M.S., Purdue (chair) **Spring 2021-Spring 2022**

Chair of the MS advisory committee. Reading course in differential privacy

Vishnu Suresh, M.S., Purdue **Spring 2021-December 2022**

PROFESSIONAL ORGANIZATIONS

Pitt Cyber: Institute for Cyber Law, Policy, and Security, Pitt
Affiliate Scholar **Spring 2026-present**

Privacy and Confidentiality Interest Group, American Statistical Association **Fall 2023-present**
Member

American Statistical Association **Summer 2017-present**

Institute of Mathematical Statistics **Summer 2017-present**

**Center for Education and Research in Information Assurance and Security (CERIAS),
Purdue** **Spring 2024-Summer 2025**

Affiliate faculty member

Regenstrief Center for Healthcare Engineering, Purdue **Summer 2022-Summer 2025**
Associate member